# L5: Diameter, Characteristic Path Length, and Network Efficiency

The notion of small-world networks depends on two concepts: how clustered the network is *(covered in the previous pages)* and how short the paths between network nodes are – that we cover next.

As you recall from Lesson-2 if a network forms a connected component we can compute the shortest-path length $d_{i,j}$ between any two nodes *i* and *j*. One way to summarize these distances for the entire network is to simply take the average such distance across all distinct node pairs. This metric *L* is referred to as **Average (shortest) Path Length (APL)** or **Characteristic Path Length (CPL)** of the network. For an undirected and connected network of n nodes, we define *L* as:

$$L = \frac{2}{n(n-1)} \sum_{i<j} d_{i,j}$$

A related metric is the harmonic mean of the shortest-path lengths across all distinct node pairs, referred to as the **efficiency** of the network:

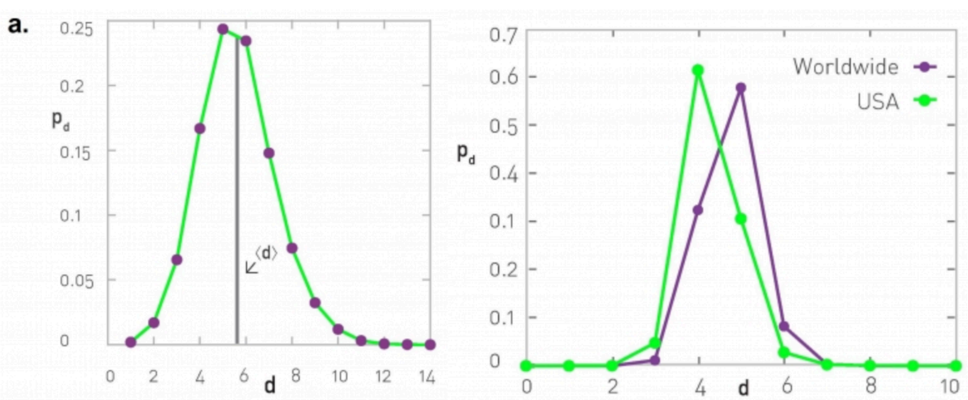
$$E = \frac{2}{n(n-1)} \sum_{i<j} \frac{1}{d_{i,j}}$$

The efficiency varies between 0 and 1.

Another metric that is often used to quantify the distance between network nodes is the **diameter**, which is defined as the maximum shortest-path distance across all node pairs:

$$D = \max_{i<j} d_{i,j}$$

A more informative description is the distribution of the shortest-path lengths across all distinct node pairs, as shown in the visualizations below.




At the left, the corresponding network is the largest connected component of the protein-protein interaction of yeast *(2,018 nodes and 2,930 edges, the largest connected component include 81% of the nodes)*. The characteristic path length *(CPL)* is 5.61 while the diameter is 14 hops.

At the right, the network is based on the *friendship* connections of Facebook users *(all pairs of Facebook users worldwide and within the US only)*. The CPL is between 4 and 5, depending on the network.